

NWL6

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NWL6 –VLANs and Trunks

V1.7

Arbeitsform:

Bildet Zweier- bzw. Dreiergruppen für den praktischen Teil.

Jede Schülerin und jeder Schüler muss aber einen **eigenständigen** Laborbericht abgeben. Duplikate bzw. weitreichende Kopien werden nicht akzeptiert. Router- und Switchnamen sind zu personalisieren. Dokumentiert am Deckblatt auch die Namen Eurer ArbeitskollegInnen.

Dauer:

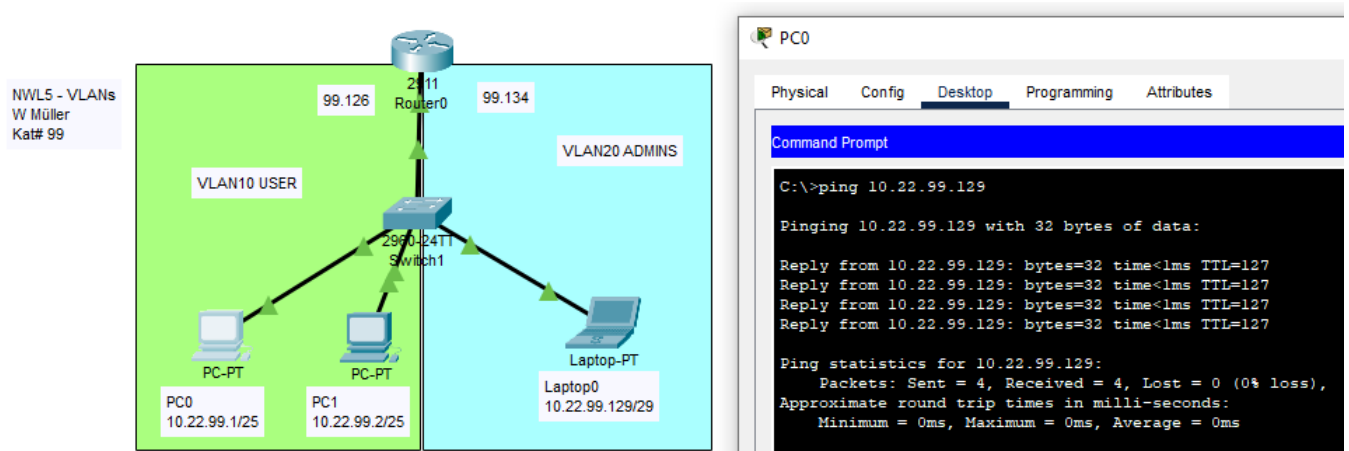
4 Schulstunden.

Hilfe:

Einführung durch Lehrer im Unterricht.

Cisco IOS Cheat-sheet

Cisco Kursunterlagen: Routing und Switching Essentials / Chapter 6 VLANs



Teil 1 – Router on a Stick - Szenario im Packettracer:

Dokumentiert und konfiguriert Euer Szenario auf Eurem persönlichen Laptop zuerst als Packettracer Szenario:

Arbeitet mit den Ergebnissen vom letzten Arbeitsauftrag weiter indem Ihr das letzte PT-Szenario als NWL6-Name erneut abspeichert und damit weiterarbeitet. Verwendet auch das Adressschema vom letzten Arbeitsauftrag!

1. Konfiguriert einen neuen 2960er Switch mit den VLANs 10_USER (/25 Netz) und 20_ADMIN (/29 Netz).
Die Ports(=Switchinterfaces) 1 bis 10 kommen ins VLAN10 , 11 bis 20 ins VLAN20.
Steckt im ersten Schritt die Leitungen vom Router (Gi0/0 bzw. Gi0/1) an das Port 10 bzw. an das Port 20, die PCs an die Ports 1,2 bzw. 11.

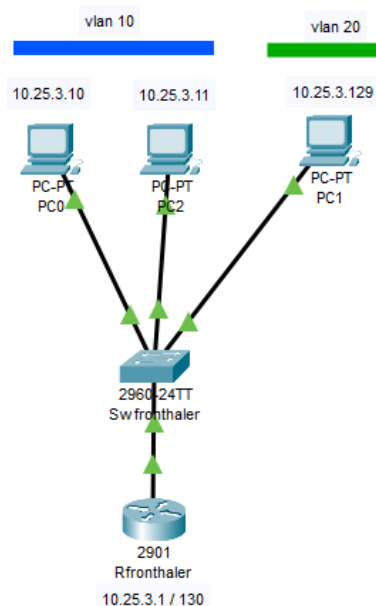
Dokumentiert die Konfigurationskommandos, das PT-Szenario sowie einen erfolgreichen ping über die Subnetzgrenze hinweg mit screenshots.

Switch config:

```
swfronthaler>enable
swfronthaler#conf term
Enter configuration commands, one per line. End with CNTL/Z.
swfronthaler(config)#vlan 10
swfronthaler(config-vlan)#name user
swfronthaler(config-vlan)#exit
swfronthaler(config)#vlan 20
swfronthaler(config-vlan)#name admin
swfronthaler(config-vlan)#exit
swfronthaler(config)#interface range fa0/1 - 10
swfronthaler(config-if-range)#switchport mode access
swfronthaler(config-if-range)#switchport access vlan 10
swfronthaler(config-if-range)#exit
swfronthaler(config)#interface range fa0/11 - 20
swfronthaler(config-if-range)#switchport mode access
swfronthaler(config-if-range)#switchport access vlan 20
swfronthaler(config-if-range)#exit
swfronthaler(config)#
```

Ports:

Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	10	--	0006.2A03.6D01
FastEthernet0/2	Down	10	--	0006.2A03.6D02
FastEthernet0/3	Up	10	--	0006.2A03.6D03
FastEthernet0/4	Down	10	--	0006.2A03.6D04
FastEthernet0/5	Down	10	--	0006.2A03.6D05
FastEthernet0/6	Down	10	--	0006.2A03.6D06
FastEthernet0/7	Down	10	--	0006.2A03.6D07
FastEthernet0/8	Down	10	--	0006.2A03.6D08
FastEthernet0/9	Down	10	--	0006.2A03.6D09
FastEthernet0/10	Down	10	--	0006.2A03.6D0A
FastEthernet0/11	Down	20	--	0006.2A03.6D0B
FastEthernet0/12	Down	20	--	0006.2A03.6D0C
FastEthernet0/13	Down	20	--	0006.2A03.6D0D
FastEthernet0/14	Down	20	--	0006.2A03.6D0E
FastEthernet0/15	Down	20	--	0006.2A03.6D0F
FastEthernet0/16	Down	20	--	0006.2A03.6D10
FastEthernet0/17	Down	20	--	0006.2A03.6D11
FastEthernet0/18	Down	20	--	0006.2A03.6D12
FastEthernet0/19	Down	20	--	0006.2A03.6D13
FastEthernet0/20	Down	20	--	0006.2A03.6D14
FastEthernet0/21	Down	1	--	0006.2A03.6D15
FastEthernet0/22	Down	1	--	0006.2A03.6D16
FastEthernet0/23	Down	1	--	0006.2A03.6D17
FastEthernet0/24	Down	1	--	0006.2A03.6D18
GigabitEthernet0/1	Down	1	--	0006.2A03.6D19
GigabitEthernet0/2	Up	1	--	0006.2A03.6D1A
Vlan1	Up	1	10.25.3.125/25	0001.C9BC.4D9E



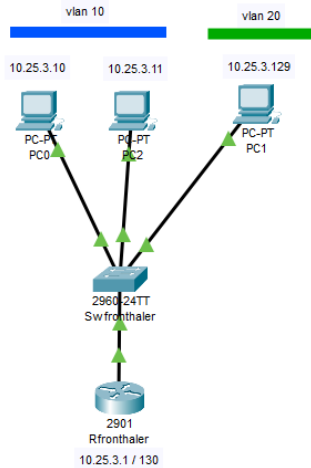
- Konfiguriert den Uplink zum Router in einen dot1q Trunk um.
Auf der Leitung werden also beide VLANs transportiert. Der Router braucht also nur mehr ein

Anschlusskabel.

Konfiguriere die IP-Subnetze am Router zu Subinterfaces um und dokumentiere die Konfigurationskommandos.

Konfiguriere am Switch am Port Gi0/1 den dot1q Uplink und erlaube dort beide VLANs.

Macht einen screenshot vom PT Szenario.



Testet erneut die Connectivity zwischen den PCs.

Pc0 auf pc2:

```
C:\>ping 10.25.3.11

Pinging 10.25.3.11 with 32 bytes of data:

Reply from 10.25.3.11: bytes=32 time<1ms TTL=128
Reply from 10.25.3.11: bytes=32 time<1ms TTL=128
Reply from 10.25.3.11: bytes=32 time<1ms TTL=128

Ping statistics for 10.25.3.11:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Pc0 auf pc1:

```
C:\>ping 10.25.3.11

Pinging 10.25.3.11 with 32 bytes of data:

Reply from 10.25.3.11: bytes=32 time<1ms TTL=127
Reply from 10.25.3.11: bytes=32 time<1ms TTL=127
Reply from 10.25.3.11: bytes=32 time<1ms TTL=127
Reply from 10.25.3.11: bytes=32 time=12ms TTL=127

Ping statistics for 10.25.3.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms
```

Merken die PCs, dass sie nun in einem VLAN hängen? Nein

Config:

```

rfronthaler>enable
rfronthaler#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
rfronthaler(config)#interface gi0/0
rfronthaler(config-if)#no ip address
rfronthaler(config-if)#no shutdown
rfronthaler(config-if)#exit
rfronthaler(config)#interface gi0/0.10
rfronthaler(config-subif)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to
up

rfronthaler(config-subif)#encapsulation dot1Q 10
rfronthaler(config-subif)#ip address 10.25.3.1 255.255.255.128
rfronthaler(config-subif)#exit
rfronthaler(config)#interface gi0/0.20
rfronthaler(config-subif)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to
up

rfronthaler(config-subif)#encapsulation dot1Q 20
rfronthaler(config-subif)#ip address 10.25.3.130 255.255.255.248
% 10.25.3.128 overlaps with GigabitEthernet0/1
rfronthaler(config-subif)#exit
rfronthaler(config)#

swfronthaler(config)#
swfronthaler(config)#interface gi0/2
swfronthaler(config-if)#switchport mode trunk

swfronthaler(config-if)#
%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to
down

%LINEPROTO-S-UPDOWN: Line protocol on Interface Vlan1, changed state to down

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to
up

%LINEPROTO-S-UPDOWN: Line protocol on Interface Vlan1, changed state to up

swfronthaler(config-if)#switchport trunk allowed vlan 10,20
swfronthaler(config-if)#

swfronthaler>enable
swfronthaler#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1
10	user	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
20	admin	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdinet-default	active	
1005	trnet-default	active	

```

swfronthaler#

```

3. Dokumentiere einen Frame auf dem Weg vom Switch zum Router und vom Router zurück zum Switch (Packet-Details im Simulation Mode am Router Incoming & Outgoing).
Ping PC0 zu PC1:

PDU Information at Device: Swfronthaler

OSI Model **Inbound PDU Details** Outbound PDU Details

PDU Formats

The screenshot displays the following details:

- Ethernet II:**
 - Preamble: 101010...10
 - Destination Address (DA): 0060.7094.178D
 - Source Address (SA): 0000.0000.0000
 - Type: 0800 (IPv4)
- IP:**
 - Version: 4
 - Header Length: 5
 - DSCP: 0x00
 - Length: 128
 - ID: 0x0014
 - Flags: 0x0
 - TTL: 127
 - Protocol: 0x01 (ICMP)
 - Checksum: 0x0000
 - Source IP: 10.25.3.10
 - Destination IP: 10.25.3.129
- ICMP:**
 - Type: 0x08 (Echo Request)
 - Code: 0x00
 - Checksum: 0x0000
 - ID: 0x000a
 - Sequence Number: 20

PDU Information at Device: Swfronthaler

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

EthernetII

PREAMBLE: 101010..10																SFD		DEST ADDR: 0060.7094.178D																															
SRC ADDR: 0060.47A1.9A01																TYPE: 0x080		DATA (VARIABLE LENGTH)																FCS: 0x00000000															

IP

VER: 4				IHL: 5				DSCP: 0x00								TL: 128																			
ID: 0x0014																FLAGS: 0x0				FRAG OFFSET: 0x000															
TTL: 127								PRO: 0x01								CHKSUM																			
SRC IP: 10.25.3.10																																			
DST IP: 10.25.3.129																																			
DATA (VARIABLE LENGTH)																																			

ICMP

TYPE: 0x08								CODE: 0x00								CHECKSUM															
ID: 0x000a																SEQ NUMBER: 20															

Variable Size PDU

DATA (VARIABLE LENGTH)																															
------------------------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

PDU Information at Device: Rfronthaler

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

Ethernet 802.1q

PREAMBLE: 101010..10																SFD		DEST ADDR: 0060.47A1.9A01																									
SRC ADDR: 0050.0FE3.0C50																TPID: 0x8100				TCI: 0x000a				Type: 0x1				DEST ADDR: 0060.47A1.9A01															
DATA (VARIABLE LENGTH)																FCS: 0x00000000																											

IP

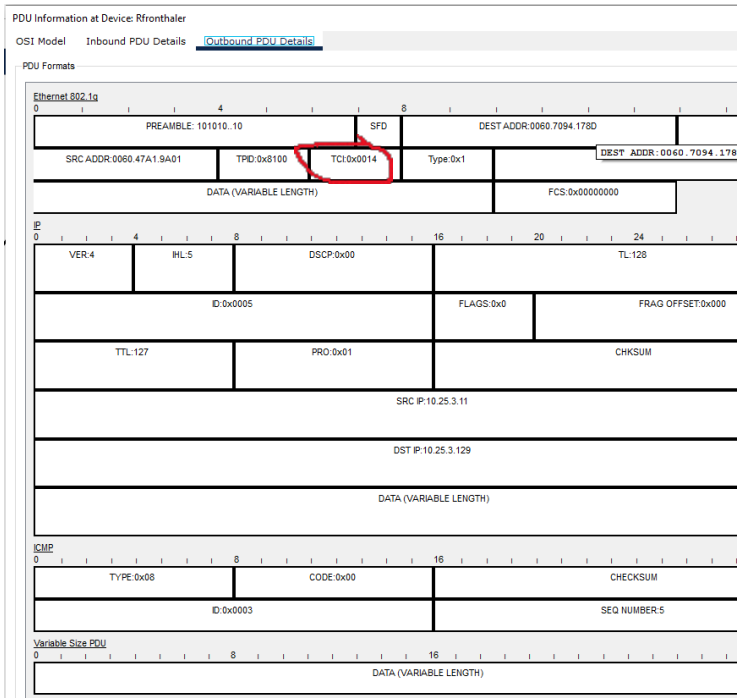
VER: 4				IHL: 5				DSCP: 0x00								TL: 128																			
ID: 0x0005																FLAGS: 0x0				FRAG															
TTL: 128								PRO: 0x01								CHKSUM																			
SRC IP: 10.25.3.11																																			
DST IP: 10.25.3.129																																			
DATA (VARIABLE LENGTH)																																			

ICMP

TYPE: 0x08								CODE: 0x00								CHECKSUM															
ID: 0x0003																SEQ NUMBER: 5															

Variable Size PDU

DATA (VARIABLE LENGTH)																															
------------------------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



Was ändert sich alles? Bei outbound ist kein 802.1q tag mehr, type ändert sich und TPID/ TCI fallen weg
 Erkennst Du das Feld mit der VLAN-ID? Ja der 802.1q header, das TCI feld ist das VLAN ID Feld.

4. Gebt die Konfiguration des Routers und Switches mit „show running-config“ aus und kopiert die wesentlichen Kommandos im Anhang des Arbeitsauftrags.

Teil 2 - Zum Nachdenken:

1. Welchen Vorteil haben VLANs und Trunks? Siehe
 Erkläre das in eigenen Worten. Wer im Dunklen tappt lese sich bitte vom letzten Jahr nochmals die Unterlagen zum „Routing und Switching Essentials“ / Chapter 6 durch.

Teil 3 - Szenario im Labor:

1. Übertragst Euer Szenario (kleinere Katalognummer des Teams) auf die physischen Geräte (Router und Switch) im Labor.
 (screenshots)

```

interface GigabitEthernet1/0/7
switchport access vlan 10
switchport mode access
!
interface GigabitEthernet1/0/8
switchport access vlan 10
switchport mode access
!
interface GigabitEthernet1/0/9
switchport access vlan 10
switchport mode access
!
interface GigabitEthernet1/0/10
switchport access vlan 10
switchport mode access
!
interface GigabitEthernet1/0/11
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet1/0/12
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet1/0/13
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet1/0/14
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet1/0/15
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet1/0/16
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet1/0/17
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet1/0/18
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet1/0/19
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet1/0/20
switchport access vlan 20
switchport mode access
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20
switchport mode trunk
!
interface GigabitEthernet1/0/25

```

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname rfalknerfronthaler
rfalknerfronthaler(config)#int GigabitEthernet 0/0
rfalknerfronthaler(config-if)#no ip address
rfalknerfronthaler(config-if)#no shutdown
rfalknerfronthaler(config-if)#int
*Jan 16 22:12:40.907: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down
*Jan 16 22:12:44.535: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jan 16 22:12:45.535: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

```



```
rfalknerfronthaler(config-subif)#int GigabitEthernet0/0.10
rfalknerfronthaler(config-subif)#encapsulation dot1Q 10
rfalknerfronthaler(config-subif)#ip address 10.25.2.126 255.255.255.128
```

```
rfalknerfronthaler(config-subif)#int GigabitEthernet0/0.20
rfalknerfronthaler(config-subif)#encapsulation dot1Q 20
rfalknerfronthaler(config-subif)#ip address 10.25.2.142 255.255.255.248
rfalknerfronthaler(config-subif)#exit
rfalknerfronthaler(config)#end
rfalknerfronthaler#write
*Jan 16 22:16:51.219: %SYS-5-CONFIG_I: Configured from console by consol
Building configuration...
[OK]
rfalknerfronthaler#write memory
Building configuration...
[OK]
rfalknerfronthaler#
```

2. Kontrolliert die IPv4 und IPv6 Connectivity mit ping zwischen den zwei PCs in den unterschiedlichen VLANs.

```
edv@nw-labor-02:~$ ping 10.25.2.126
PING 10.25.2.126 (10.25.2.126) 56(84) bytes of data.
64 bytes from 10.25.2.126: icmp_seq=1 ttl=255 time=0.514 ms
64 bytes from 10.25.2.126: icmp_seq=2 ttl=255 time=0.594 ms
^C
--- 10.25.2.126 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1020ms
rtt min/avg/max/mdev = 0.514/0.554/0.594/0.040 ms
edv@nw-labor-02:~$ ping 10.25.2.142
PING 10.25.2.142 (10.25.2.142) 56(84) bytes of data.
64 bytes from 10.25.2.142: icmp_seq=1 ttl=255 time=0.576 ms
64 bytes from 10.25.2.142: icmp_seq=2 ttl=255 time=0.531 ms
64 bytes from 10.25.2.142: icmp_seq=3 ttl=255 time=0.507 ms
^C
--- 10.25.2.142 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2028ms
rtt min/avg/max/mdev = 0.507/0.538/0.576/0.028 ms
edv@nw-labor-02:~$ ping 10.25.2.137
PING 10.25.2.137 (10.25.2.137) 56(84) bytes of data.
64 bytes from 10.25.2.137: icmp_seq=1 ttl=63 time=1.22 ms
64 bytes from 10.25.2.137: icmp_seq=2 ttl=63 time=0.673 ms
^C
--- 10.25.2.137 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.673/0.946/1.220/0.273 ms
edv@nw-labor-02:~$
```

(screenshots)

```

rtr min/avg/max/mdev = 0.498/0.499/0.499/0.499 ms
edv@nw-labor-02:~$ ping FE80::7281:5FF:FEF5:C980
PING FE80::7281:5FF:FEF5:C980(fe80::7281:5ff:fef5:c980) 56 data bytes
64 bytes from fe80::7281:5ff:fef5:c980%en0: icmp_seq=1 ttl=64 time=3.81 ms
64 bytes from fe80::7281:5ff:fef5:c980%en0: icmp_seq=2 ttl=64 time=0.508 ms
64 bytes from fe80::7281:5ff:fef5:c980%en0: icmp_seq=3 ttl=64 time=0.500 ms
64 bytes from fe80::7281:5ff:fef5:c980%en0: icmp_seq=4 ttl=64 time=0.535 ms
^C
--- FE80::7281:5FF:FEF5:C980 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3028ms
rtt min/avg/max/mdev = 0.500/1.338/3.810/1.427 ms
edv@nw-labor-02:~$ ping 2001:db8:2:10::1
PING 2001:db8:2:10::1(2001:db8:2:10::1) 56 data bytes
64 bytes from 2001:db8:2:10::1: icmp_seq=1 ttl=64 time=0.592 ms
64 bytes from 2001:db8:2:10::1: icmp_seq=2 ttl=64 time=0.513 ms
64 bytes from 2001:db8:2:10::1: icmp_seq=3 ttl=64 time=0.524 ms
^C
--- 2001:db8:2:10::1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2028ms
rtt min/avg/max/mdev = 0.513/0.543/0.592/0.034 ms

```

That's it, Bravo!

Bitte wieder als NWL6-*Name*.PDF im moodle abgeben!

Anhang: Konfiguration des Switches und des Routers (wesentliche Teile)

! Switchkonfiguration

Building configuration...

Current configuration : 2310 bytes

!

version 15.0

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname swfronthaler

!

!

!

ip ssh version 2

ip domain-name fronthaler.local

!

```
username admin secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
```

```
!
```

```
!
```

```
!
```

```
spanning-tree mode pvst
```

```
spanning-tree extend system-id
```

```
!
```

```
interface FastEthernet0/1
```

```
switchport access vlan 10
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/2
```

```
switchport access vlan 10
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/3
```

```
switchport access vlan 10
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/4
```

```
switchport access vlan 10
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/5
```

```
switchport access vlan 10
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/6
```

```
switchport access vlan 10
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/7
```

```
switchport access vlan 10
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/8
```

```
switchport access vlan 10
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/9
```

```
switchport access vlan 10
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/10
```

```
switchport access vlan 10
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/11
```

```
switchport access vlan 20
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/12
```

```
switchport access vlan 20
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/13
```

```
switchport access vlan 20
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/14
```

```
switchport access vlan 20
```

```
switchport mode access
```

```
!
```

```
interface FastEthernet0/15
```

```
switchport access vlan 20
```

```
switchport mode access
!
interface FastEthernet0/16
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/17
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/18
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/19
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/20
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
switchport mode trunk
!
interface GigabitEthernet0/2
switchport trunk allowed vlan 10,20
switchport mode trunk
!
interface Vlan1
ip address 10.25.3.125 255.255.255.128
!
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
!
end
```

! Routerkonfiguration

Building configuration...

Current configuration : 1045 bytes

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

```
!  
hostname rfronthaler  
!  
!  
!  
!  
!  
!  
!  
!  
!  
ip cef  
ipv6 unicast-routing  
!  
no ipv6 cef  
!  
!  
!  
username admin secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1  
!  
!  
license udi pid CISCO2901/K9 sn FTX1524602I-  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
ip ssh version 2  
ip domain-name rfronthaler.local  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
description to_switch  
no ip address  
duplex auto  
speed auto  
ipv6 address 2001:DB8:AC03::1/64  
!  
interface GigabitEthernet0/0.10  
encapsulation dot1Q 10  
ip address 10.25.3.1 255.255.255.128  
!  
interface GigabitEthernet0/0.20  
encapsulation dot1Q 20  
ip address 10.25.3.130 255.255.255.248  
!  
interface GigabitEthernet0/1  
description to_pc  
no ip address  
duplex auto  
speed auto  
ipv6 address 2001:DB8:AC03:1::1/64  
shutdown  
!  
interface Vlan1
```

```
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login local
!
!
!
end
```